Rojae Mighty

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Education

Stony Brook University *Expected Graduation: 05/27*
B.S. in Physics (Natural Sciences)
Programming languages: **Python**, **C/C++** · Relevant coursework: statistical mechanics, computational methods, data analysis, probability

Research Experience

Brookhaven National Laboratory *2025–Present*

- **Department of Energy Science Undergraduate Laboratory Internship (SULI)** (Summers 2025, 2026): Electron-ion collision studies of semi-inclusive processes; implemented a spatially dependent parton distribution function using **ROOT**, **Python**, and **Unix/Linux shell scripting**.

Center for Frontiers in Nuclear Science (CFNS) *2025–Present*
Undergraduate Researcher *Advisor: Zhoudunming (Kong) Tu*

- Developed **machine learning** pipelines in **Python** with **scikit-learn** for kinematic reconstruction on large-scale collision datasets; applied **data science** methods including **feature engineering** and **statistical data analysis**.
- Used **C++** and **ROOT** with **Monte Carlo event generators** to study gluon saturation through geometric scaling; ran **numerical simulation** studies of high-energy nuclear collision phenomena.
- Delivered **conference presentations** at DIS 2026, APS Global Physics Summit (oral), and Stony Brook Physics Colloquium; prepared materials in **LaTeX**.

Projects

Market Microstructure Forecasting *2025–Present* *GitHub*

- Built a **Python machine learning** pipeline with **pandas**, **NumPy**, and **scikit-learn**; **feature engineering** and **walk-forward validation** on time-series limit order book data; **Git**-versioned on GitHub.

J.P. Morgan Quantitative Research (Forage) *2026*

- Forage certificate: commodity **statistical data analysis**; **derivatives pricing**; **Python** credit-risk model; FICO bucketing via **dynamic programming/scikit-learn** (MSE/log-likelihood).

Honors & Presentations

- 1st Place, Physical Sciences Presentation, Yale Undergraduate Research Conference (2026)
- Outstanding Oral Presentation Award, National Society of Black Physicists Conference (2025)
- Barry Barish Undergraduate Research Travel Award, Stony Brook University (2026)
- *Geometric Scaling and Gluon Saturation at the EIC* — DIS 2026, Yale URC, NSBP, APS DNP (CEU), APS Global Physics Summit, Stony Brook Colloquium

Technical Skills

Languages:	Python, C/C++, Unix/Linux shell scripting
Libraries:	pandas, NumPy, scikit-learn
ML & data:	machine learning, data science, statistical data analysis, feature engineering, walk-forward validation, numerical simulation, dynamic programming
Tools:	ROOT, Git, LaTeX, Monte Carlo event generators